Sean Summers Assignment 6.2

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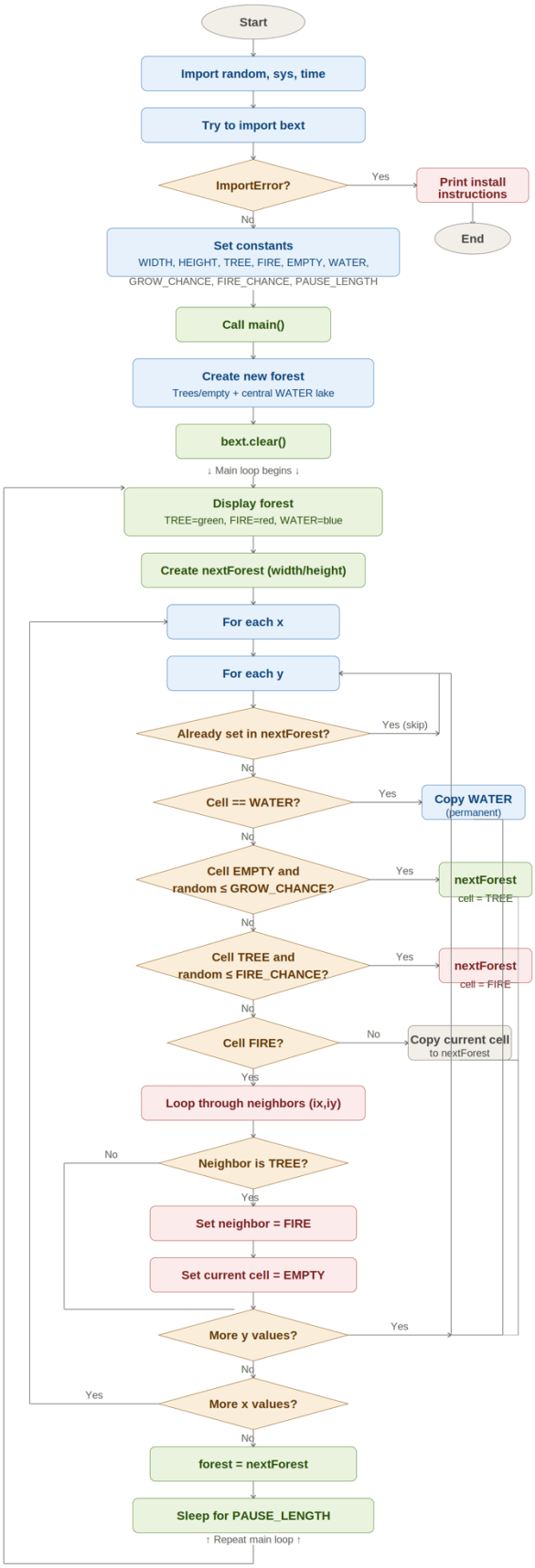
Start
Import random, sys, time
Try to import bext
ImportError?
Yes
Print install instructions
End
No
Set constants
WIDTH, HEIGHT, TREE, FIRE, EMPTY, WATER,
GROW_CHANCE, FIRE_CHANCE, PAUSE_LENGTH
Call main()
Create new forest
Trees/empty + central WATER lake
bext.clear()
↓ Main loop begins ↓
Display forest
TREE=green, FIRE=red, WATER=blue
Create nextForest (width/height)
For each x
For each y
Already set in nextForest?
Yes (skip)
No
Cell == WATER?
Yes
Copy WATER
(permanent)
No
Cell EMPTY and
random ≤ GROW_CHANCE?
Yes
nextForest
cell = TREE
No
Cell TREE and
random ≤ FIRE_CHANCE?
Yes
nextForest
cell = FIRE
No
Cell FIRE?
No
Copy current cell
to nextForest
Yes
Loop through neighbors (ix,iy)
Neighbor is TREE?
No
Yes
Set neighbor = FIRE
Set current cell = EMPTY
More y values?
Yes
No
More x values?
Yes
No
forest = nextForest
Sleep for PAUSE_LENGTH
↑ Repeat main loop ↑